SS-C-540B
June 11, 1975
SUPERSEDING
Fed. Spec. SS-C-540A
March 1, 1967

FEDERAL SPECIFICATION

COAL TAR (CUTBACK) ROOF COATING, BRUSHING CONSISTENCY

This specification was approved by the Commissioner, Federal Supply Service, General Services Administration, for the use of all Federal agencies.

1. SCOPE AND CLASSIFICATION

1.1 Scope. This specification covers a coal tar base roof coating containing mineral filler or stabilizer and volatile solvent, suitable for brush application.

1.2 Classification. Coal tar shall be of one type only.

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issues in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

Federal Specifications:

HH-R-595 - Roofing Felt, Coat Tar and Asphalt Saturated Organic Felts, Rolls.
PPP-D-705 - Drum: Metal Shipping, Steel, (Over 12 and Under 55 Gallon).
PPP-D-729 - Drums, Shipping and Storage, Steel, 55-Gallon.
PPP-D-732 - Drums, Metal, 55-Gallon Reconditioned (For Shipment of Non-corrosive Material).
PPP-P-704 - Pails, Metal: (Shipping, Steel, 1 through 12 Gallons).

Federal Standard:

Fed. Std. No. 123 - Marking for Shipment (Civil Agencies).

(Activities outside the Federal Government may obtain copies of Federal Specifications, Standards, and Handbooks as outlined under General Information in the Index of Federal Specifications and Standards and at the prices indicated in the Index. The Index, which includes cumulative monthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, U.S. Government Printing Office, Washington, DC 20402.

(Single copies of this specification and other Federal Specifications required by activities outside the Federal Government for bidding purposes are available without charge from Business Service Centers at the General Services Administration Regional Offices in Boston, New York, Washington, DC, Atlanta, Chicago, Kansas City, MO, Fort Worth, Denver, San Francisco, Los Angeles, and Seattle, WA.

(Federal Government activities may obtain copies of Federal Specifications, Standards, and Handbooks and the Index of Federal Specifications and Standards from established distribution points in their agencies.)

Military Standards:

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129 - Marking for Shipment and Storage.
MIL-STD-147 - Palletized and Containerized Unit Loads 40 Inch x 48 Inch Pallets, Skids, Runners or Pallet Type Base.

(Copies of Military Specifications and Standards required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

FSC 5610

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2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless a specific issue is identified, the issue in effect on date of invitation for bids or request for proposal shall apply.

American Society for Testing and Materials (ASTM) Standards:

D 93 - Flash Point by Pensky-Martens Closed Tester.
D 95 - Water in Petroleum Products and Bituminous Materials by Distillation.
D 482 - Ash from Petroleum Products.

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

National Motor Freight Traffic Association, Inc., Agent:

National Motor Freight Classification.

(Application for copies should be addressed to the American Trucking Associations, Inc., Tariff Order Section, 1616 P Street, N.W., Washington, DC 20036.)

Uniform Classification Committee, Agent:

Uniform Freight Classification.

(Application for copies should be addressed to the Uniform Classification Committee, Room 1106, 222 South Riverside Plaza., Chicago, IL 60606.)

3. REQUIREMENTS

3.1 Material. The coating shall consist of refined coke overtar, plasticizing oils, and aromatic and/or aliphatic tar solvents blended with mineral fillers (usually asbestos fiber) or stabilizers to yield a heavy brushing consistency ready for use.

3.1.1 Nonvolatile material. The roof coating shall yield not less than 60 percent nonvolatile matter, when tested in accordance with 4.4.1.

3.1.2 Ash content. The material shall have an ash content of not less than 5 percent, and not more than 12 percent of a suitable mineral fiber or stabilizer, when tested in accordance with 4.4.2.

3.1.3 Water content. The roof coating shall have a maximum water content of 0.5 percent by weight of coating, when tested in accordance with 4.4.3.

3.2 Condition in container. The material shall be a smooth, homogeneous mixture after stirring and shall be ready for use without heating or thinning. Settling in the container shall not occur to the extent that a hard cake which cannot be redispersed by hand mixing forms at the bottom of the container at 60°F or above.

3.3 Working properties.

3.3.1 Coating. Coating shall be of a heavy brushing consistency to insure a film thickness of approximately 1/32 inch at material temperature up to 90°F (32.2°C), and yet not so stiff at 60°F (15.5°C) to prevent easy application with a suitable brush at the rate of approximately 1 gallon per 50 square feet over bituminous roofing, when tested in accordance with 4.4.4.

3.3.2 Viscosity. The Brookfield viscosity of the coating at 77° ± 2°F (25° ± 1.1°C) shall be not less than 6, nor more than 16 when tested in accordance with 4.4.5.

3.4 Flashpoint. The flashpoint of the coating shall be a minimum of 170°F (76.6°C) when tested in accordance with 4.4.6.

3.5 Penetration and saturation. The coating shall penetrate and stain at least the top 12 blotter sheets in the stack, and at least the top 9 blotter sheets shall show saturation a minimum of 95 percent of the total surface area of each sheet. The twelfth blotter sheet from the top shall show a stain area of at least 2 inches in diameter, when tested in accordance with 4.4.7.

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3.6 Behavior at warm and cold temperatures.

3.6.1 Accelerated aging. The coating shall not become hard or brittle, crack, or lose adhesion, when tested in accordance with 4.4.8.1.

3.6.2 Low temperature flexibility. The coating shall show no more than hairline surface cracks and no loss of adhesion, when tested in accordance with 4.4.8.2.

3.7 Shelf storage life. When stored as specified in 4.4.9, the coating shall conform to 3.1.3, 3.2, and 3.3.2. However, the viscosity may vary ± 10 percent from the values specified in 3.3.2.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements.

4.2 Sampling.

4.2.1 Lot. Unless otherwise specified (see 6.2), a lot shall consist of all coating submitted for delivery at the same time.

4.2.2 Sampling for tests. Sampling for tests shall be in accordance with MIL-STD-105, with level S-4 for 1 and 5 gallon cans, and level S-2 for 30 and 55 gallon drums. The sample unit shall be one container. Failure of any test specified in paragraph 4.4 shall be cause for rejection of the lot.

4.3 Examination.

4.3.1 End item. Coating selected in accordance with 4.2.2 shall be examined for defects listed in table I. The acceptable quality levels (AQL's) for defects shall be 2.5 percent defective for major defects, and 4.0 percent for minor defects in accordance with MIL-STD-105.

TABLE I. Classification of defects, end item

Defects	Major	Minor
Coating not ready for use without heating or thinning	X	
Hard cakes in containers which are not dispersed by hand mixing	X	
Damage or defects not affecting functions or serviceability		X
Damage or defects affecting function or serviceability	X	

4.3.2 Inspection of preparation for delivery. An inspection shall be made to determine that the packaging, packing, and marking requirements comply with section 5. Defects shall be scored in accordance with table II. The lot size shall be the number of shipping containers fully prepared for delivery, with the exception that containers need not be sealed or closed, nor interior containers or case liners sealed (if applicable). Examination shall be in two phases; first an interior examination in process of packaging, and second an examination of containers fully prepared for delivery. The sample unit for each of the two phases shall be a container prepared for delivery as set forth above. The inspection level shall be S-2 of MIL-STD-105, with an AQL of 4.0 percent defective.

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TABLE II. Classification of preparation for delivery defects

Examine	Defects
Marking (interior package or container and exterior container as applicable)	Omitted, incorrect, illegible, improper size, location, sequence, or method of application.
Materials	Component missing, damaged, defective.
Workmanship (as applicable)	Inadequate or improper packaging or packing, such as closure of interior packages or containers, closure of case liners or container flaps, taping of seams, corners, and manufacturer's joint, closure of alternate containers; loose strapping or tape banding; inadequate stapling, bulging or distortion of containers.
Contents (interior and exterior containers as applicable)	Number per container not as specified.
Weight (exterior containers)	Weight per container exceeds maximum specified.
Preservation (as applicable)	Preservative missing, improperly applied or incorrect type.

4.4 Tests.

4.4.1 Nonvolatile matter. Weigh approximately 10 grams (g.) of the sample into a tarred, flat bottom metal dish about 8 cm. in diameter, or into a quart friction top can plug. Heat the dish with its contents in a forced draft circulating oven at 325° ± 2°F (162.8° ± 1.1°C) for five hours. Cool and weigh. From the weight of the residue in the dish and from the weight of the original sample taken, compute the percentage of nonvolatile matter, (see 3.1.1).

4.4.2 Ash content. Scrape from the metal dish in 4.4.1 a large portion of the residue nonvolatile matter into a previously fired and tarred crucible. Weigh the sample. Slowly burn off the combustible matter and ignite to constant weight. Cool and weigh. Calculate the percent ash based on the original sample taken for the nonvolatile matter. As an alternate method for percent ash use procedure for ash in ASTM D 482 (see 3.1.2).

4.4.3 Water content. Determine water content in accordance with ASTM D 95, (see 3.1.3).

4.4.4 Coating. Prepare an area 50 by 36 inches of roofing felt conforming to type I, class 2 of HH-R-595, firmly fastened and in a horizontal position. Apply to this specimen 1 quart of the coating material at a temperature of 73.5° ± 2°F (23.1° ± 1.1°C). Spread the coating over the surface, using a suitable brush which has been saturated with the coating material (see 3.3.1).

4.4.5 Viscosity. Determine the viscosity of the coating on a model HBT 200 or equivalent Brookfield Viscosimeter using a #3 Spindle at 2.5 RPM at 77° ± 2°F (25° ± 1.1°C). Read values on the 0-100 scale. The sample tested should be contained in a one pint can, (see 3.3.2).

4.4.6 Flashpoint. Determine flashpoint in accordance with ASTM D 93 using the procedure for determination of flashpoint of suspension of solids, (see 3.4).

4.4.7 Penetration and saturation. Place a metal ring 2-5/8 inches in diameter and 1/2-inch high on top of a stack of 20 sheets of 4 by 4 inch white blotter paper. (Blotter specifications white, uncoated 200M/1000 sheets). Stir the coating, until homogeneous, and fill the ring level full with the coating. Place the ring-papercoating assembly in a forced draft circulating oven for 24 hours at 158° ± 2°F (70° ± 1.1°C). Remove the assembly from the oven, and cool to room temperature of 73.5 ± 2°F (23.1 ± 1.1°C). Carefully, remove the ring from the assembly by running a thin blade knife around the inside of the ring separating the coating from the ring. Examine the blotter paper stack for depth of penetration of coating and saturation of sheets (see 3.5).

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4.4.8 Behavior at warm and cold temperature.

4.4.8.1 Accelerated aging. Prepare a stack of 20 pieces of 40 Whatman filter paper, 9 cm. in diameter. Place the stack of filter papers on one sheet of 4 inch by 4 inch blotter paper. Place a ring as in 4.4.7 on top of the stack, and fill level full with a sample of thoroughly mixed coating. Age the test assembly in a forced-draft circulating oven at 158° ± 2°F (70° ± 1.1°C) for seven days. Remove the test assembly from the oven and cool to room temperature of 73.5° ± 2°F (23.1° ± 1.1°C). As in 4.4.7, remove the ring from the assembly. Bend the specimen over 1/4-inch mandrel in not more than two seconds. Examine for cracking, loss of adhesion, embrittlement, (see 3.6.1).

4.4.8.2 Low temperature flexibility. After recording data for 4.4.7, place the paper-coating assembly into a cold box at 25°F for six hours. Remove the assembly from the cold box and bend over a 1/4-inch mandrel within two seconds. Not more than five seconds should elapse between removal of specimen from cold box and bending of specimen (see 3.6.2).

4.4.9 Shelf storage life. One container of coating shall be stored for 18 months from the date of manufacture at 73.5° ± 2°F (23.1° ± 1.1°C) and a relative humidity of 50 ± 5 percent and then shall be tested to determine compliance with the requirements of 3.7.

5. PREPARATION FOR DELIVERY

5.1 Packaging. Packaging shall be level A, B, or C, as specified (see 6.2).

5.1.1 Level A. The coal tar shall be packaged in 1 gallon or 5 gallon pails conforming to PPP-P-704, type II, class 3; 30 gallon drums conforming to PPP-D-705, type IV or V, or 55 gallon drums conforming to PPP-D-729, type III or IV, or PPP-D-732, type III or IV, class B, as specified (see 6.2).

5.1.2 Level B. The coal tar furnished in 1, 5, 30 or 55 gallon quantities shall be packaged as specified in 5.1.1.

5.1.3 Level C. The coal tar shall be packaged in accordance with the supplier's commercial practice.

5.2 Packing. Packing shall be level A, B, or C, as specified (see 6.2).

5.2.1 Levels A and B. The coal tar packaged in 1 gallon and 5 gallon pails shall be palletized in accordance with MIL-STD-147, load, type IV. The 30 gallon drums and the 55 gallon drums require no overpacking.

5.2.2 Level C. The coal tar shall be packed to insure carrier acceptance and safe delivery to destination at lowest rates in containers complying with the Uniform Freight Classification Rules or National Motor Freight Classification Rules.

5.3 Marking.

5.3.1 Civil agencies. In addition to markings required by the contract or order, the pails and drums shall be marked in accordance with Fed. Std. No. 123.

5.3.2 Military agencies. In addition to markings required by the contract or order, the pails and drums shall be marked in accordance with MIL-STD-129.

6. NOTES

6.1 Intended use. Coal tar roof coating is intended for use in the repair and coating of tar base roofing and for treating concrete construction to resist water.

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6.2 Ordering data. Purchasers should select the preferred options permitted herein, and include the following information in procurement documents:

(a) Title, number, and date of this specification.
(b) Size of lot, if different from 4.2.1.
(c) Selection of applicable level of packaging and packing required (see 5.1 and 5.2).
(d) Size of container required (see 5.1.1).

MILITARY INTERESTS:

Custodian:

Army - ME

User activities:

Army - ME
Navy - YD
Air Force - 84

Preparing Activity:

GSA-FSS

CIVIL AGENCY COORDINATING ACTIVITY:

COM-NBS

Orders for this publication are to be placed with General Services Administration, acting as an agent for the Superintendent of Documents. See Section 2 of this specification to obtain extra copies and other documents referenced herein. Price 30 cents each.

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